MULTI-VARIABLE - II
[CALCULUS]

Expected Ques-Ans [2023-24]

Q1. $\underset{(x,y)\to(0,0)}{Lt} \dfrac{x}{\sqrt{x^2+y^2}}$ is

→ put $Y = mx + c$

at $(x,y) \to (0,0)$

$c = 0$

$Y = mx$

$$\underset{(x,mx)\to(0,0)}{Lt} \frac{x}{\sqrt{x^2+m^2x^2}}$$

$$\underset{x\to 0}{Lt} \frac{x}{\sqrt{x^2(1+m^2)}}$$

$$\underset{x\to 0}{Lt} \frac{x}{x\sqrt{1+m^2}}$$

$$\underset{x\to 0}{Lt} \frac{1}{\sqrt{1+m^2}}$$

Hence, $\dfrac{1}{\sqrt{1+m^2}}$ is independent from x

$\therefore$ Limit does not exists

Q.2. The value of

$$\underset{(x,y)\to(k,0)}{Lt}\left[1+\frac{x}{y}\right]^{y} \text{ is}$$

$\Rightarrow$ put $Y = mx + c$

at $(x,y)\to(k,0)$

$0 = mk + c$

$c = -mk$

$Y = mx - mk$

$Y = m(x-k)$

$$\underset{x\to k}{Lt}\left[1+\frac{x}{m(x-k)}\right]^{m(x-k)} \quad [\infty^{0} \text{ form}]$$

let $$z = \underset{x\to k}{Lt}\left[1+\frac{x}{m(x-k)}\right]^{m(x-k)}$$

$$\log z = \underset{x\to k}{Lt}\log\left[1+\frac{x}{m(x-k)}\right]^{m(x-k)}$$

$$\log z = \underset{x\to k}{Lt}\, m(x-k)\cdot\log\left[1+\frac{x}{m(x-k)}\right]$$

$$\log z = \underset{x\to k}{Lt}\left[\frac{\log(1+x/m(x-k))}{\frac{1}{m(x-k)}}\right] \quad \left[\frac{0}{\infty} \text{ form}\right]$$

Applying L' hospital's rule

$$\log z = \underset{x \to k}{Lt} \frac{\left[\dfrac{1}{1 + \dfrac{x}{m(x-k)}}\right] \times \dfrac{m(x-k) - mx}{[m(x-k)]^2}}{\dfrac{-1}{[m(x-k)]^2}(m)}$$

$$\log z = \underset{x \to k}{Lt} \frac{\left[\dfrac{1}{1 + \dfrac{x}{m(x-k)}}\right] \times \dfrac{-mk}{[m(x-k)]^2}}{\dfrac{-m}{[m(x-k)]^2}}$$

$$\log z = \underset{x \to k}{Lt} \frac{\cancel{[m(x-k)]^2}(-mk)}{\left[1 + \dfrac{x}{m(x-k)}\right]\cancel{[m(x-k)]^2}(-m)}$$

$$\log z = \underset{x \to k}{Lt} \frac{-mk}{(-m)\left[1 + \dfrac{x}{m(x-k)}\right]}$$

$$\log z = \underset{x \to k}{Lt} \frac{k}{\left[1 + \dfrac{x}{m(x-k)}\right]}$$

$$\log z = 0$$

$\log z = 0$

$e^{\log z} = e^0$

$\boxed{z = 1}$ $\qquad [e^0 = 1]$

Q.3. Discuss continuity of the following function at the point $(0,0)$:

$$f(x,y) = \begin{cases} \left(\dfrac{x^2+y^2}{xy}\right) & ; (x,y) \neq (0,0) \\ 0 & ; (x,y) = (0,0) \end{cases}$$

$\Rightarrow$ Given, $f(x,y) = \dfrac{x^2+y^2}{xy}$

$f(0,0) = 0$

To check continuity

$$\underset{\substack{x \to a \\ y \to b}}{Lt} f(x,y) = f(a,b)$$

Now, $\underset{\substack{x \to 0 \\ y \to 0}}{Lt} \dfrac{x^2+y^2}{xy}$

put $\begin{cases} y = mx + c \\ \text{at } (x,y) \to (0,0) \end{cases}$

$c = 0$

$y = mx$

$$\lim_{x \to 0} \frac{x^2 + m^2x^2}{x m x}$$

$$= \lim_{x \to 0} \frac{x^2[1+m^2]}{m x^2}$$

$$= \lim_{x \to 0} \frac{1+m^2}{m}$$

$$= \frac{1+m^2}{m}$$

$$\lim_{\substack{x \to 0 \\ y \to 0}} f(x,y) = \frac{1+m^2}{m}$$

$f(0,0) = 0$

Hence, $\lim_{\substack{x \to 0 \\ y \to 0}} f(x,y) \neq f(0,0)$

$\therefore$ function is ~~discontinu~~ discontinuous at origin.

Q.4. Find the extreme value of x^2+y^2, when $x^4+y^4=1$

$\Rightarrow$

~~Given;~~

~~let $f(x) = x^2+y^2$~~

~~$g(x) = x^4+y^4-1$~~

~~Now,~~

~~$F = f(x) + \lambda g(x)$~~

~~$= x^2+y^2+\lambda(x^4+y^4-1)$~~

~~Here~~

$\Rightarrow$ Given;

let $f(x,y) = x^2+y^2$

$g(x,y) = x^4+y^4-1$

Now,

$F = \cancel{f(x)+\lambda}\ f(x,y) + \lambda g(x,y)$

$F = x^2+y^2+\lambda(x^4+y^4-1)$

Here,

$$\frac{\partial F}{\partial x} = 2x + 4\lambda x^3$$

$$\frac{\partial F}{\partial y} = 2y + 4\lambda y^3$$

Now,

$$\frac{\partial F}{\partial x} = 0 \quad \text{and} \quad \frac{\partial F}{\partial y} = 0$$

$$2x + 4\lambda x^3 = 0 \quad \left[\text{from } \frac{\partial F}{\partial x} = 0\right]$$

$$4\lambda x^3 = -2x$$

$$\lambda = \frac{-2x}{4x^3}$$

$$\lambda = -\frac{1}{2x^2} \quad \text{—— (i)}$$

$$2y + 4\lambda y^3 = 0 \quad \left[\text{from } \frac{\partial F}{\partial x} = 0\right]$$

$$2y = -4\lambda y^3$$

$$\lambda = \frac{-2y}{4y^3}$$

$$\lambda = -\frac{1}{2y^2} \quad \text{—— (ii)}$$

Solving eq- (i) and (ii), we have

$$\frac{-1}{2x^2} = \frac{-1}{2y^2}$$

$$x^2 = y^2 \quad \text{—— (iii)}$$

$x^4 + y^4 = 1$ [as given]

$(x^2)^2 + (y^2)^2 = 1$

$(x^2)^2 + (x^2)^2 = 1$ [from eq-(iii)]

$2x^4 = 1$

$x^4 = \frac{1}{2}$

$x = \frac{1}{(2)^{1/4}}$

$Y = \frac{1}{(2)^{1/4}}$ [from eq- iii]

$\therefore$ extreme value of $x^2 + y^2$

$= \left(\frac{1}{2^{1/4}}\right)^2 + \left(\frac{1}{(2)^{1/4}}\right)^2$

$= \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}}$

$= \frac{2}{\sqrt{2}} = \sqrt{2}$

Hence, extreme value of x^2+y^2 is $\sqrt{2}$.

Q.5 Discuss the continuity of the following function at the point (0,0):

$$f(x,y) = \begin{cases} \dfrac{x^2y^2}{(x^3+y^3)} & , (x,y) \neq (0,0) \\ 0 & , (x,y) = (0,0) \end{cases}$$

⇒ Given;

$$f(x,y) = \frac{x^2y^2}{x^3+y^3}$$

$$f(0,0) = 0$$

Now; $f(x,y) = \dfrac{x^2y^2}{x^3+y^3}$ — (i)

Here, $y = mx + c$

at $(x,y) \to (0,0)$

$c = 0$

$y = mx$

Put the value of y in eq — (i)

$$f(x,mx) = \frac{x^2 m^2 x^2}{x^3 + m^3x^3}$$

$$f(x,mx) = \frac{\cancel{x^4}^{\,x}\, m^2}{\cancel{x^3}(1+m^3)}$$

$$f(x,mx) = \frac{xm^2}{1+m^3}$$

$$\lim_{x \to 0} f(x, mx) = \lim_{x \to 0} \frac{x m^2}{1+m^2}$$

$$\lim_{x \to 0} f(x, mx) = 0$$

Hence, $f(0,0) = 0$

$$\lim_{x \to 0} f(x,y) = 0$$

Hence, $f(0,0) = \lim_{x \to 0} f(x,y)$

$\therefore$ Then function is continuous.

Q.6. Find the maximum value of xyz under the constraints $x^2+z^2=1$ and $y-x=0$.

$\Rightarrow$ Given;

$$f(x,y,z) = xyz$$

$$g(x,y,z) = x^2+z^2-1$$

$$h(x,y,z) = y-x$$

Now, $F = f(x,y,z) + \lambda_1 g(x,y,z) + \lambda_2 h(x,y,z)$

$$F = xyz + \lambda_1 (x^2+z^2-1) + \lambda_2 (y-x)$$

Here,

$$\frac{\partial F}{\partial x} = yz + 2\lambda_1 x - \lambda_2$$

$$\frac{\partial F}{\partial y} = xz + \lambda_2$$

$$\frac{\partial F}{\partial z} = xy + 2\lambda_1 z$$

Now,

Solving for $\frac{\partial F}{\partial x} = 0$, $\frac{\partial F}{\partial y} = 0$ and $\frac{\partial F}{\partial z} = 0$

$yz + 2\lambda_1 x - \lambda_2 = 0$ [from $\frac{\partial F}{\partial x} = 0$] — (i)

$xz + \lambda_2 = 0$ [from $\frac{\partial F}{\partial y} = 0$]

$\lambda_2 = -xz$ — (ii)

$xy + 2\lambda_1 z = 0$ [from $\frac{\partial F}{\partial z} = 0$]

$\lambda_1 = -\frac{xy}{2z}$ — (iii)

Now, putting the value of λ_1, λ_2 in eq-(i)

$$yz + 2\left(-\frac{xy}{2z}\right) + xz = 0$$

$$yz - \frac{xy}{z} + xz = 0$$

$$yz^2 - xy + xz^2 = 0$$

$\cancel{z^2(y-x) - xy = 0}$

$\cancel{xy = 0}$ $\quad$ $\cancel{[(y-x=0) \rightarrow \text{given}]}$

$\cancel{x = 0 \quad \text{or} \quad y = 0}$

$xy = z^2(x+y)$ $\quad$ $\left[\begin{array}{l} y - x = 0 \text{ given} \\ \Rightarrow y = x \end{array}\right]$

$y^2 = z^2(2y)$

$y^2 = 2z^2$

$x^2 = 2z^2$

Now putting $x = 2z^2$ in $x^2 + z^2 = 1$

$2z^2 + z^2 = 1$

$3z^2 = 1$

$z^2 = \frac{1}{3}$

$z = \frac{1}{\sqrt{3}}$

$x^2 = 2z^2$

$x = \sqrt{\frac{2}{3}}$

$y^2 = 2z^2$

$y = \sqrt{\frac{2}{3}}$

Hence, maximum value of xyz

$$= \sqrt{\frac{2}{3}} \times \sqrt{\frac{2}{3}} \times \frac{1}{\sqrt{3}} = \frac{2}{3\sqrt{3}}$$

Q.7. Compute $\frac{\partial^2 f}{\partial x \partial y}(0,0)$ and $\frac{\partial^2 f}{\partial y \partial x}(0,0)$ for the function

$$f(x,y) = \begin{cases} \frac{xy^3}{x+y^2}, & (x,y) \neq (0,0) \\ 0, & (x,y) = (0,0) \end{cases}$$

Also discuss the continuities of $\frac{\partial^2 f}{\partial x \partial y}$ and $\frac{\partial^2 f}{\partial y \partial x}$ at $(0,0)$.

$\Rightarrow$ Given;

$$f(x,y) = \frac{xy^3}{x+y^2}$$

$$f(0,0) = 0$$

first order partial derivative w.r.t x

Now, $$\frac{\partial f}{\partial x} = \frac{y^3[x+y^2] - xy^3}{(x+y^2)^2}$$

$$\frac{\partial f}{\partial x} = \frac{xy^3 + y^5 - xy^3}{(x+y^2)^2}$$

$$\frac{\partial f}{\partial x} = \frac{y^5}{(x+y^2)^2}$$

first order partial derivative w.r.t y

$$\frac{\partial f}{\partial y} = \frac{3xy^2[x+y^2] - xy^3[2y]}{(x+y^2)^2}$$

$$\frac{\partial f}{\partial y} = \frac{3x^2y^2 + 3xy^4 - 2xy^4}{(x+y^2)^2}$$

$$\frac{\partial f}{\partial y} = \frac{3x^2y^2 + xy^4}{(x+y^2)^2}$$

Mixed order partial derivative of x w.r.t y

$$\left\{\frac{\partial}{\partial y}\left(\frac{\partial f}{\partial x}\right)\right\}$$

$$\Rightarrow \frac{\partial^2 f}{\partial y \partial x} = \frac{\partial}{\partial y}\left[\frac{y^5}{(x+y^2)^2}\right]$$

$$= \frac{5y^4[x+y^2]^2 - y^5 2(x+y^2)\cdot 2y}{(x+y^2)^4}$$

$$= \frac{5y^4[x+y^2]^2 - 4y^6[x+y^2]}{(x+y^2)^4}$$

$$= \frac{\cancel{(x+y^2)}\left[5y^4[x+y^2] - 4y^6\right]}{\cancel{(x+y^2)^4}\,(x+y^2)^3}$$

$$= \frac{5xy^4 + 5y^6 - 4y^6}{(x+y^2)^3}$$

$$\frac{\partial^2 f}{\partial y \partial x} = \frac{5xy^4 + y^6}{(x+y^2)^3} \quad \text{--- (4)}$$

Mixed order partial derivative of y w.r.t x

$$\frac{\partial^2 f}{\partial x \partial y} = \frac{\partial}{\partial x}\left[\frac{\partial f}{\partial y}\right]$$

$$= \frac{[y^4 + 6xy^2][x+y^2]^2 - [3x^2y^2 + xy^4]\,2(x+y^2)}{(x+y^2)^4}$$

$$= \frac{\cancel{(x+y^2)}\,[y^4 + 6xy^2(x+y^2)] - 2[3x^2y^2 + xy^4]}{\cancel{(x+y^2)^4}\,(x+y^2)^3}$$

$$= \frac{xy^4 + y^6 + \cancel{6x^2y^2} - \cancel{6x^2y^2} - 2xy^4 + 6xy^4}{(x+y^2)^3}$$

$$\frac{\partial^2 f}{\partial x \partial y} = \frac{5xy^4 + y^6}{(x+y^2)^3} \quad \text{— (ii)}$$

solving eq-(i) and (ii) we have

$$\frac{\partial^2 f}{\partial x \partial y} = \frac{\partial^2 f}{\partial y \partial x}$$

Now, checking continuity of $\frac{\partial^2 f}{\partial x \partial y}$ and $\frac{\partial^2 f}{\partial y \partial x}$

As, we know $\frac{\partial^2 f}{\partial x \partial y} = \frac{\partial^2 f}{\partial y \partial x}$

$$\therefore \frac{\partial^2 f}{\partial x \partial y} = \frac{5xy^4 + y^6}{(x+y^2)^3}$$

Here, $y = mx + c$

$$(x, y) \to (0, 0)$$

$$c = 0$$

$$y = mx$$

putting $y = mx$ in $\frac{\partial^2 F}{\partial x \partial y}$, we have

$$\frac{\partial^2 F}{\partial x \partial y} = \frac{5x m^4 x^4 + m^6 x^6}{(x + m^2 x^2)^3}$$

$$\frac{\partial^2 F}{\partial x \partial y} = \frac{x^5 [5m^4 + x m^6]}{x^3 [1 + mx]^3}$$

$$\frac{\partial^2 F}{\partial x \partial y} = \frac{x^2 [5m^4 + x m^6]}{(1 + mx)^3}$$

$$\underset{\substack{x \to 0 \\ y \to 0}}{Lt} \frac{\partial^2 F}{\partial x \partial y} = \underset{\substack{x \to 0 \\ y \to 0}}{Lt} \frac{x^2 [5x^4 + x m^6]}{[1 + mx]^3}$$

$$= 0$$

Hence, $f(0,0) = \underset{\substack{x \to 0 \\ y \to 0}}{Lt} f(x, y)$

Hence function is continuous.

Q8. Find the minimum value of $x^2+y^2+z^2$ subject to constraint $xyz=a^3$.

=> Given:

let $f(x,y,z) = x^2+y^2+z^2$

let $g(x,y,z) = xyz - a^3$

Now,

$$F = f(x,y,z) + \lambda(g(x,y,z))$$

$$F = x^2+y^2+z^2+\lambda(xyz-a^3)$$

Here

$$\frac{\partial F}{\partial x} = 2x + \lambda yz$$

$$\frac{\partial F}{\partial y} = 2y + \lambda xz$$

$$\frac{\partial F}{\partial z} = 2z + \lambda xy$$

Solving, $\frac{\partial F}{\partial x} = 0$, $\frac{\partial F}{\partial y} = 0$ and $\frac{\partial F}{\partial z} = 0$

$$2x + \lambda yz = 0 \quad \left(\text{from } \frac{\partial F}{\partial x} = 0\right)$$

$$\lambda = -\frac{2x}{yz} \quad \text{— (1)}$$

$$2y + \lambda xz = 0 \quad \left(\text{from } \frac{\partial F}{\partial y} = 0\right)$$

$$\lambda = \frac{-2y}{xz} \quad \text{— (2)}$$

$$2z + \lambda xy = 0 \quad \left(\text{from } \frac{\partial f}{\partial z} = 0\right)$$

$$\lambda = \frac{-2z}{xy} \quad \text{— (3)}$$

from eq (1) (2) (3), we get

$$\frac{-2x}{yz} = \frac{-2y}{xz} = \frac{-2z}{xy}$$

$$x^2 z = y^2 z \Rightarrow x^2 = y^2$$

$$xy^2 = xz^2 \Rightarrow y^2 = z^2$$

$$\therefore x^2 = y^2 = z^2$$

$$x = y = z \quad \text{— (4)}$$

As given in ques.

$$xyz = a^3$$

$$x^3 = a^3 \quad (\text{from (4)})$$

$$x = a$$

$$y = a$$

$$z = a$$

maximum value of $x^2+y^2+z^2$ is

$$a^2+a^2+a^2 = 3a^2$$

Hence maximum value of $x^2+y^2+z^2 = 3a^2$

Q.9 Find the equation of the tangent plane to the surface $x^2-3y^2-z^2=2$, at the point $(3,1,2)$.

⇒ Given;

let $f(x,y,z) = x^2-3y^2-z^2-2$

Here,

$$\frac{\partial F}{\partial x} = 2x$$

$$\frac{\partial F}{\partial y} = -6y$$

$$\frac{\partial F}{\partial z} = -2z$$

Now, $\frac{\partial F}{\partial x}$, $\frac{\partial F}{\partial y}$ and $\frac{\partial F}{\partial z}$ at the point $(3,1,2)$

$$\frac{\partial F}{\partial x} = 6$$

$$\frac{\partial F}{\partial y} = -6$$

$$\frac{\partial F}{\partial z} = -4$$

Equation of tangent plane to the surface at the point (3,1,2);

$$(X-x)\frac{\partial F}{\partial x} + (Y-y)\frac{\partial F}{\partial y} + (Z-z)\frac{\partial F}{\partial z} = 0$$

$$(X-3)\,6 + (Y-1)(-6) + (Z-2)(-4) = 0$$

$$6X - 18 - 6Y + 6 - 4z + 8 = 0$$

$$6X - 6Y - 4z - 4 = 0$$

$$3X - 3Y - 2z - 2 = 0$$

Hence, equation of tangent is

$$3X - 3Y - 2z - 2 = 0.$$

Q.10.) Find the equation of normal to the surface

$$z^2 = 4(1+x^2+y^2) \text{ at } (2,2,6)$$

$\Rightarrow$ Given;

$$F(x,y,z) = 4(1+x^2+y^2) - z^2$$

Here, $\frac{\partial F}{\partial x} = 8x$

$\frac{\partial F}{\partial y} = 8y$

$$\frac{\partial f}{\partial z} = -2z$$

Now, $\frac{\partial f}{\partial x}$, $\frac{\partial f}{\partial y}$ and $\frac{\partial f}{\partial z}$ at $(2,2,6)$

$$\frac{\partial f}{\partial x} = 16$$

$$\frac{\partial f}{\partial y} = 16$$

$$\frac{\partial f}{\partial z} = -12$$

Equation of normal to the surface at the point $(2,2,6)$;

$$\frac{(X-x)}{\frac{\partial f}{\partial x}} = \frac{Y-y}{\frac{\partial f}{\partial y}} = \frac{Z-z}{\frac{\partial f}{\partial z}}$$

$$\frac{X-2}{16} = \frac{y-2}{16} = \frac{z-6}{-12}$$

$$\frac{X-2}{4} = \frac{y-2}{4} = \frac{z-6}{-3}$$

$$\frac{X-2}{4} = \frac{y-2}{4} = \frac{6-z}{3}$$

Hence, this is equation normal to the surface.